```
anim.UpdateNodeDescription(wifiStaNodes.Get(0), "Wireless_
N1");
anim.UpdateNodecolour(wifiStaNodes.Get(0), 255, 0, 0);
```

Execute the following code in a terminal opened in the directory *ns/ns-allinone-3.22/ns-3.22* to run the ns-3 script *wireless.cc*.

```
./waf
./waf -- run scratch/wireless
```

On execution, the terminal will display some log information, which I have saved in a text file titled *LOG_INFORMATION.txt* and this file is also included in the downloadable Zip file. Figure 1 shows the *NetAnim* window running the animation trace file *wireless.xml* generated by *wireless.cc*.

Plotting graphs with ns-3 data

The saying 'A picture paints a thousand words' is true even in the field of scientific research. The results presented by a thousand page treatise can often be summarised as an elegant graph. Your research will be followed more widely and appreciated if you could represent the results in the form of a graph. The very popular plotting tool called *gnuplot* is used for plotting graphs from ns-3 data. Keep in mind that even though the name of this tool starts with 'gnu' it doesn't have anything to do with the GNU Project and neither does it support the GNU GPL. Both 2-dimensional and 3-dimensional plots are possible with *gnuplot*. The function *plot* is used for plotting 2D graphs and the function *splot* is used for plotting 3D graphs. The input for the 2D plot is a file containing two-column numeric data and the input for the 3D plot is a file containing three-column numeric data. So everything boils down to the process of obtaining the data file that can be given as an input to the plot functions of *gnuplot*.

There are two different ways to do this in ns-3. In the first method, the role of ns-3 is just to produce the trace file. Afterwards, this trace file is processed by some Linux commands or text processing utilities like AWK or Perl to extract two-column or three-column numeric data, as required. This data is then given as an input to *gnuplot*. This method is simple if you know how to extract data with Linux commands or text processing utilities, in which case, no additional knowledge of ns-3 is required.

The code below shows the Linux command cut being used to extract two-column data from the trace file *wireless.tr* generated by *wireless.cc*. The extracted two-column data is redirected to a file called *plot_input*. This file is then used by *gnuplot* to plot the graph.

```
# cut -d' ' -f2,28 wireless.tr > plot_input
# gnuplot
gnuplot> set xlabel "Time"
gnuplot> set ylabel "Packet Size"
gnuplot> set title "Graph"
gnuplot> plot "plot_input"
gnuplot> exit
```

The option *–d* of the *cut* command tells which character is used as a field delimiter for deciding individual columns. Here, the delimiter is whitespace. An instance of the *gnuplot* utility is invoked with the command *gnuplot*. The commands *set xlabel* and *set ylabel* are used to set the X-axis and Y-axis labels. The title of the graph is given by the command *settitle*. Since we are dealing with two-column data, the 2D function *plot* is used. The name of the input file should be enclosed within double quotes. With this we will obtain a graph but the use of Linux commands may not always give us all the required data to produce a graph. In such situations, we might be forced to use AWK or Perl scripts.

The previous method of plotting graphs depends heavily on our knowledge of Linux commands and text processing utilities, and the data is always extracted from the trace file. What if we need to access data directly from various ns-3 modules? In order to access data directly from the source to plot a graph, ns-3 provides a class called *Gnuplot*. The code below shows an ns-3 script called *graph.cc* using the *Gnuplot* class.

```
#include "ns3/gnuplot.h"
#include <fstream>
#include <math.h>
using namespace ns3;
using namespace std;
int main (int argc, char *argv[])
{
  double x,y,t;
  Gnuplot plot("graph.eps");
  plot.SetTitle("2-D Graph");
```

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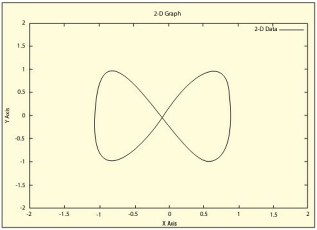

Figure 2: Graph plotted with gnuplot